

The exact evaluation

The Birch–Swinnerton-Dyer finite sum for $y^2 = x^3 + px$, computed, and what it turns the Fermat pattern into

fourth note on III at the Fermat primes

1. The formula

The previous note ended by saying that what was wanted was an **exact evaluation**, not a lower bound, of the finite sum that expresses the central L -value. Here it is, and here is what it gives.

For a CM elliptic curve the central value is a finite sum of special values of the Weierstrass function — Birch and Swinnerton-Dyer’s expression, restated as Theorem 2.6 of [Nomoto](#). Our curves fall in its first branch, because $(i/p)_4 = i^{(p-1)/2} = 1$ for every $p \equiv 1 \pmod{8}$. Taking $D = \Delta = p$:

$$\frac{\varepsilon p}{\Omega} L_{2p}(\bar{\psi}_{-p}, 1) = \frac{\sqrt{2}}{4} \sum_c \left(\frac{c}{p}\right)_4 + \frac{1}{\sqrt{2}} \sum_c \left(\frac{c}{p}\right)_4 \frac{\mathcal{P}_c + 1}{\mathcal{P}_c^2 - 2\mathcal{P}_c - 1},$$

the sums being over $c \in (\mathcal{O}_K/p)^\times$, where $\mathcal{P}_c = \mathcal{P}(c\Omega/p)$ is the Weierstrass function of the lattice $\Omega\mathcal{O}_K$ — so $\mathcal{P}_c$ is the x -coordinate of a p -torsion point of $E_1 : y^2 = x^3 - x$ — and $\Omega = 2.6220575\dots$

The first sum **vanishes**: $(\cdot/p)_4$ is a non-trivial character and c runs over the whole group. So everything is in the second sum, which we call $S(p)$. Note that its denominator $w^2 - 2w - 1$ has roots $1 \pm \sqrt{2} = \mathcal{P}(\Omega/4), \mathcal{P}((1+2i)\Omega/4)$: the poles sit at the 4-torsion.

Normalisation check. Before trusting anything, `sha-bsdsum.gp` reproduces the paper’s own special values to 60 digits: $\Omega = 2.62205755\dots$, $\mathcal{P}(\Omega/4) = 1 + \sqrt{2}$, $\mathcal{P}'(\Omega/4) = -4 - 2\sqrt{2}$, $\mathcal{P}((1+2i)\Omega/4) = 1 - \sqrt{2}$.

2. The evaluation

Computing $S(p)$ to 60 digits and recognising it with `algdep`:

$$S(17)^4 = 2^6 \cdot 17^3, \quad S(257)^4 = 2^{14} \cdot 257^3.$$

That is, $S(17) = 2^{3/2} \cdot 17^{3/4}$ and $S(257) = 2^{7/2} \cdot 257^{3/4}$ — **exact** algebraic numbers, with no odd part whatever.

The factor $p^{3/4}$ is not a surprise: the period of $y^2 = x^3 + px$ is $\Omega/p^{1/4}$, the quartic twist scaling the period, and the theorem’s left-hand side carries Ω rather than the twisted period. Dividing it out and comparing with the quantity m of the second note, defined by $L(E_p, 1)/\Omega_E = 2m^2$:

The finite sum computes m^2 on the nose. On every prime tested, to 40 digits,

$$S(p) = 2\sqrt{2} \cdot m^2 \cdot p^{3/4}.$$

p	$S(p)/p^{3/4}$	$S(p)/(2\sqrt{2}p^{3/4})$	m
17	$2\sqrt{2}$	1	1
41	$2\sqrt{2}$	1	1
97	$2\sqrt{2}$	1	1
137	$2\sqrt{2}$	1	1
257	$4 \cdot 2\sqrt{2}$	4	2
577	$4 \cdot 2\sqrt{2}$	4	2

3. What the pattern has become

Putting the four notes together, the Fermat-prime observation is now this, and nothing else:

For $p = 2^{2^k} + 1$, the character sum over the p -torsion of $y^2 = x^3 - x$

$$\sum_{c \in (\mathcal{O}_K/p)^\times} \left(\frac{c}{p} \right)_4 \frac{\mathcal{P}(c\Omega/p) + 1}{\mathcal{P}(c\Omega/p)^2 - 2\mathcal{P}(c\Omega/p) - 1}$$

equals $2\sqrt{2} \cdot 4^{k-2} \cdot p^{3/4}$.

Everything else is done: this value gives $m = 2^{k-2}$, hence $L(1)/\Omega = 2^{2k-3}$, hence $\#\text{III} = 2^{2k-2}$, and the shape $(\mathbb{Z}/2^{k-1})^2$ was already a theorem. The whole mystery is now one identity about one finite sum.

4. Why this is the right target, and what to try

Three features make the reformulation worth having.

It is finite and elementary. No L -functions, no BSD: a sum of $(p-1)^2$ algebraic numbers, each a rational function of the x -coordinate of a torsion point, weighted by a quartic residue symbol.

It explains the square. m^2 appears, not m — the sum is intrinsically a square, which is why $\#\text{III} = (2m)^2$ and why the Cassels pairing had to be alternating. The two came out consistent because they are the same fact.

And it is a Gauss-sum-shaped object. A sum over a finite group, weighted by a multiplicative character of 2-power order, of values of a fixed algebraic function. The tool for the **exact** 2-adic valuation of such sums is *Stickelberger's theorem*, whose answer is a **sum of base- p digits**. Digit sums are exactly the sort of quantity that is generically of size $\sqrt{p}$ and collapses to something tiny on a thin family — and, crucially, they can behave doubly logarithmically along a family like $p-1 = 2^{2^k}$, which is precisely the growth $\log_2 m = \log_2 \log_2 b - 1$ that no algebraic formula in a and b could produce (second note). That is where we would look next.

The obstacle to more data. $S(p)$ has $(p-1)^2$ terms, so $p = 65537$ needs $4.3 \cdot 10^9$ evaluations of $\mathcal{P}$ — out of reach this way, even though the analytic L -value for that p was computed in the first note. Any further progress needs the sum evaluated **2-adically** rather than numerically: reduce the torsion points and the character into a 2-adic field and compute the valuation directly. That is the concrete next step, and it is the step Zhao's method takes by induction on prime factors — an induction that, as the third note showed, has nothing to induct on when D is a single split prime.

5. Status

Done here. The finite sum is implemented, its normalisation validated against four independent special values, and evaluated exactly for $p = 17$ and 257 : $S = 2^{3/2}p^{3/4}$ and $2^{7/2}p^{3/4}$. The identity $S = 2\sqrt{2}m^2p^{3/4}$ is confirmed on six primes to 40 digits, so the sum computes m^2 directly and the Fermat pattern is reduced to a single explicit identity.

Not done. The identity itself. What remains is a 2-adic evaluation of a quartic-character sum over torsion points — Stickelberger territory — rather than anything about elliptic curves.